

DeSurv identifies a replicable tumor–stroma prognostic architecture in pancreatic cancer through survival-supervised matrix factorization

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) outcomes reflect malignant, stromal, and other microenvironmental biology. Expression-driven discovery can overlook prognostically relevant programs that contribute relatively little transcriptomic variation. We developed DeSurv, a survival-supervised matrix factorization that incorporates patient outcomes into gene-program discovery and freezes gene weights for new cohorts. In pooled TCGA and CPTAC cohorts, DeSurv recovered a three-program tumor–stroma architecture whose dominant survival-aligned program combined classical-tumor and restCAF-like stromal features. Applied without refitting to five external datasets, the frozen risk score remained prognostic before (HR per SD, 1.42; 95% CI, 1.26–1.59) and after PurIST and DeCAF adjustment. Unsupervised factorization matched on program number and tuning budget did not recover this program; a rank-optimized unsupervised control recovered comparable prognostic information across seven factors. In simulations with known underlying structure, DeSurv recovered prognostic genes more accurately when survival signal contributed little expression variation. Survival supervision therefore offers a route to prognostically relevant, interpretable programs.

Pancreatic ductal adenocarcinoma (PDAC) is among the most lethal solid tumors¹. Its clinical heterogeneity reflects variation in both malignant cells and the surrounding microenvironment. Bulk transcriptomic studies have identified clinically relevant malignant and stromal states in PDAC: basal-like and classical tumor programs^{2–4}, and, from single-cell-informed analyses, tumor-promoting proCAF and tumor-restraining restCAF stromal programs⁵. Single-cell and spatial profiling have further shown that this heterogeneity extends across malignant, stromal, immune, and other microenvironmental compartments within individual tumors^{6–8}. Expression-defined programs can be evaluated against survival after discovery^{2,9}. Whether allowing survival to shape program definitions can identify a tumor–stroma program that provides prognostic information beyond established classifications remains unclear. Such programs could provide candidates for

prognostic assays and trial stratification, while identifying genes and biological processes for further investigation. The question is not specific to PDAC, since in any tumor type the largest sources of expression variation need not be those most relevant to survival.

Addressing this question requires large cohorts with survival-annotated transcriptomic data, which in PDAC are still derived primarily from bulk profiling¹⁰. Bulk profiles, however, mix signals from malignant and microenvironmental cells¹¹. Nonnegative matrix factorization (NMF) and related deconvolution methods separate these composite profiles into interpretable gene programs, defined as weighted sets of co-varying genes that together form a representation of the tumor transcriptome^{3,12–14}. In PDAC, these approaches have supported molecular subtype discovery and subsequent single-sample classification^{2,4,9}, virtual microdissection of tumor- and stroma-specific expression³, and compartment-specific tumor–stroma characterization^{5,14}. When predefined signatures are applied unchanged to a bulk cohort, survival can be used to assess their prognostic relevance but does not alter their gene definitions during that application. The practical challenge is therefore not simply to resolve interpretable gene programs. It is to let survival shape the programs as they are learned, to retain a decomposed representation rather than only a risk score, and to test those same programs, unchanged, across independent studies.

Standard NMF learns gene programs through a reconstruction-driven objective, minimizing reconstruction error, defined as the mismatch between the measured expression matrix and its approximation rebuilt from the programs. Such reconstruction-driven criteria favor patterns that capture prominent expression variation regardless of their relevance to patient survival^{15,16}. This is particularly important in PDAC, where bulk tumors contain substantial stromal, exocrine, and other non-malignant signals that can dominate transcriptomic variation and obscure less prominent but prognostically relevant programs^{4,5,17,18}.

Reconstruction alone also does not uniquely identify the gene programs without additional constraints^{19–21}, which may contribute to cross-study variability among proposed unsupervised PDAC subtype taxonomies²². Establishing clinically relevant and transportable programs has involved extensive multi-study refinement: the basal-like/classical dichotomy developed over roughly a decade of iterative subtyping, virtual microdissection, independent cohorts, single-cell analyses and prospective clinical profiling^{2–4,8,10}, and comparable work on stromal CAF subtypes is under way⁵. Survival information offers an additional criterion for shaping discovery, although it does not by itself resolve this non-uniqueness.

Survival-supervised models such as penalized Cox regression and supervised principal components identify outcome-associated expression patterns, but do not explicitly separate those patterns into malignant and microenvironmental programs^{15,23}. PDAC subtypes and their single-sample classifiers, however, are defined at the program level^{4,5}.

Here, we present DeSurv, a survival-supervised matrix factorization framework that integrates NMF with Cox regression, moving clinical relevance from a post hoc criterion to one that shapes gene programs during discovery. Because the factorization is sensitive to initialization, solutions from many independent restarts are aggregated to initialize a final fit. The resulting gene weights are then held fixed, or frozen, so the same learned programs can be tested in external cohorts without refitting the representation.

We evaluate DeSurv in PDAC by comparing the resulting representation with unsupervised NMF at matched and higher rank (the number of programs the factorization extracts) and with conventional survival models such as penalized Cox regression and supervised principal components (Supplementary Table S1). We then evaluate the clinical relevance of the frozen programs in independent

cohorts and ask whether they retain prognostic information beyond established PDAC classifiers. Simulations in which the true underlying (latent) programs are known characterize the conditions under which survival supervision improves recovery of prognostic structure. Together, these analyses evaluate survival-guided program discovery in PDAC and examine its potential when prognostically relevant structure contributes relatively little expression variation.

Results

DeSurv directly incorporates survival information into gene program discovery

DeSurv learns gene programs by balancing reconstruction of the tumor transcriptome with association to patient survival, so that outcome information shapes which programs are recovered rather than only which programs are selected after an unsupervised factorization. A supervision parameter α controls the strength of survival supervision, with $\alpha = 0$ leaving a purely reconstruction-driven factorization, and the rank k sets how many programs are extracted (Fig. 1a; Methods). After training, the gene weights and the Cox coefficients are fixed, so the same learned programs and the trained risk score apply unchanged to new tumors without refitting the factorization.

We trained DeSurv in pooled TCGA and CPTAC PDAC cohorts^{24,25} ($n = 273, 139$ events). Hyperparameters were selected by Bayesian optimization^{26,27}, a sequential search algorithm that proposes each new setting from the results of previous ones to optimize a pre-specified objective, here cross-validated concordance (the C-index, the probability that the model ranks a pair of patients in the correct survival order; Methods). Concordance favored intermediate survival supervision over the unsupervised $\alpha = 0$ boundary but changed little across ranks (Supplementary Figs. S1 and S4). We therefore applied the one-standard-error rule, which takes the most parsimonious model within one standard error of the best, selecting $k = 3$; the remaining hyperparameters, including $\alpha = 0.334$, were retained from the overall best configuration, which had rank $k = 7$ (Fig. 1b; Methods; Supplementary Information, Bayesian Optimization). The reported combination was therefore never itself evaluated. In contrast, standard unsupervised rank criteria (reconstruction residuals, cophenetic correlation and silhouette width^{13,28}) gave conflicting rank choices in the same data (Supplementary Fig. S5). All model and hyperparameter selection was completed within the training cohorts.

The final fit was initialized from a consensus of 100 random restarts (Methods). Individual restarts reached similar training concordance (median C-index 0.70, range 0.68–0.73) yet dispersed gene sets (median pairwise Jaccard of top-ranked genes 0.13 against 0.07 by chance), so the consensus step, not any single restart, defines the reported programs. Consensus fits from each half of 8 split-half partitions of that restart pool agreed closely with the reported basis (median gene-loading correlations 0.97–0.99, median top-ranked-gene Jaccard 0.73–0.88; Supplementary Information, Stability of the recovered gene programs). These partitions assess sensitivity to restart-pool composition within the same training data; they do not independently repeat the full discovery procedure.

Survival supervision changes which gene programs are recovered

To compare the recovered gene programs at the same rank, we fitted DeSurv and standard Lee multiplicative NMF with $k = 3$. DeSurv resolved three distinct programs (D1–D3): D1, enriched for classical tumor and restCAF-like stromal signatures; D2, enriched for proCAF-like stroma; and D3,

enriched for basal-like tumor programs (Fig. 2a). Together, these programs define a tumor–stroma architecture organized around the survival-aligned D1 direction. In the COMPASS trial cohort, D1 scores correlated with GATA6 RNA in situ hybridization, independently supporting the classical tumor component of D1 (Spearman $\rho = 0.68$, $P < 0.001$, $n = 33$; Fig. 2e).

Standard NMF, the unsupervised comparator, recovered a different set of programs (Fig. 2b). Its malignant factor (N1) was enriched for both classical and basal-like tumor programs, its microenvironmental factor (N3) for multiple stromal and immune programs^{29,30}, and a third factor (N2) for exocrine-associated variation. Both methods recovered the activated-stroma and proCAF-like reference programs but differed at the tumor-restraining end: among the displayed top-ranked genes, the restCAF and iCAF reference programs corresponded to a DeSurv program but to no standard NMF factor (Fig. 2a,b). These are bulk gene-program correspondences, not identifications of CAF cell states (Supplementary Information, Stromal resolution).

This difference reflected selective reorganization rather than wholesale loss of expression structure. D1 accounted for only 18.8% of the full model’s reconstruction gain (Shapley shares³¹), the smallest share among the three programs. However, in post-fit unpenalized Cox models of the three training-cohort scores projected over all analyzed genes, omitting D1 and refitting the Cox model using the remaining scores lowered the maximized Cox partial log-likelihood, the fit criterion of the Cox model, by 66.4, compared with 1.0 for D3 and 0.02 for D2 (Fig. 2c). These conditional, nonadditive increments describe associations in the training data used for discovery and do not independently validate prognostic relevance. Thus, the programs that best reconstruct the transcriptome need not be those most relevant to survival¹⁸.

The principal NMF tumor and microenvironmental factors corresponded closely to D3 ($\rho = 0.84$) and D2 ($\rho = 0.87$), whereas D1 correlated only weakly with every matched-rank NMF factor ($\rho \leq 0.36$; gene-loading Spearman correlation over all analyzed genes; Fig. 2d). DeSurv reconstructed the analysis matrix less completely than matched-rank NMF (R^2 0.75 versus 0.81, a difference of 6.0 percentage points; Supplementary Table S6). Survival supervision therefore preserved most of the variance-dominant tumor and stromal structure while resolving a distinct survival-aligned program.

Tuning effort did not appear to explain this difference. We repeated the comparison with a matched-budget unsupervised control, which used DeSurv’s optimizer, gene filter, cross-validation folds, Bayesian-optimization budget and consensus initialization, with rank fixed at three, α fixed at zero and the remaining hyperparameters re-optimized (Methods). The control again recovered D2 ($\rho = 0.86$) and D3 ($\rho = 0.75$) but similarly failed to recover D1 ($\rho \leq 0.24$).

Frozen programs retain prognostic relevance across independent PDAC cohorts, with exploratory transport to treated disease

We next projected the frozen programs into five external validation datasets from four independent, non-metastatic, treatment-naïve PDAC cohorts (Dijk, Moffitt, PACA-AU [array and RNA-seq], and Puleo; Supplementary Table S2). Each tumor was scored by projecting its expression onto the trained programs over the union of their top-ranked genes (810 genes; $Z = \widetilde{W}^\top X_{\text{new}}$; Methods). Combined analyses represented each patient once, yielding 570 unique patients and 388 deaths (Methods).

D1 had protective point estimates in all five validation datasets and was the only individual DeSurv program with a statistically significant combined association in the dataset-stratified analysis (HR per within-dataset SD 0.75, 95% CI 0.68–0.83, $P < 0.001$). D3 had hazard ratios above 1 in four

of the five datasets (1.09, 95% CI 0.98–1.21, $P = 0.10$), while D2 was near-null overall (1.02, 95% CI 0.92–1.12, $P = 0.76$) and varied more across datasets; neither reached statistical significance individually or in combination (Fig. 3a). Nor did any of the three standard NMF factors at the same rank (HRs 0.94–1.00; Fig. 3a, right).

The trained DeSurv linear predictor combines D1 and D3, with coefficients oriented so that higher values indicate higher risk; the elastic-net penalty set the D2 coefficient to exactly zero. It was associated with overall survival in the dataset-stratified analysis (HR per pooled SD 1.42, 95% CI 1.26–1.59, $P = 5.4 \times 10^{-9}$; Table 1), and leave-one-dataset-out estimates were similar (HRs 1.34–1.45; Fig. 3a). Scaled Schoenfeld residuals indicated attenuation over follow-up ($\chi^2 = 14.7$, 1 df, $P = 1.3 \times 10^{-4}$; Supplementary Information, Proportional-hazards diagnostic), so it should be read as a time-averaged summary rather than a constant multiplicative effect.

Allowing cohort-specific slopes did not significantly improve fit over a common slope (likelihood-ratio $\chi^2 = 1.5$, 3 df, $P = 0.69$; score standardized within each of the four cohorts). Discrimination across cohorts was moderate (event-weighted C-index 0.61, 95% CI 0.58–0.65). In a sensitivity analysis recalibrating the frozen score in sample within each validation dataset (apparent rather than out-of-sample error), the integrated Brier score, an event-weighted summary over dataset-specific follow-up horizons, was 0.159 compared with 0.168 for a Kaplan–Meier reference (Supplementary Information).

We also projected the frozen programs, without retraining, into two treated PDAC cohorts. In the O’Kane/COMPASS cohort of metastatic disease ($n = 173$, 123 deaths), D1 remained associated with overall survival in a Cox model stratified by treatment arm (HR per SD 0.75, 95% CI 0.62–0.90; Fig. 4a); these laser-capture microdissected profiles are epithelial-enriched, probing chiefly the tumor component of D1. In the Linehan cohort, the projected proCAF-like D2 score increased from pre- to post-treatment biopsies (mean change +0.34 SD, paired Wilcoxon $P = 0.013$; Fig. 4b; Supplementary Information, Translational transportability), a shift that may reflect cell composition, cellular state or both.

Table 1: Pooled external validation of DeSurv and unsupervised controls. Hazard ratios per pooled SD of the linear predictor across five validation datasets spanning four independent cohorts (570 unique patients, 388 deaths). Adjusted models include PurIST and DeCAF and are stratified by dataset; all 570 patients had both classifier calls. The matched-budget control fixes rank at $k = 3$, whereas the rank-optimized control uses the rank selected by the same one-standard-error rule applied to its own Bayesian-optimization results ($k = 7$); both use the same fitting pipeline, consensus procedure and outcome-based tuning budget as DeSurv, with α fixed at zero (Supplementary Table S8). These are point estimates for the selected models, not tests of superiority and not a comparison of minimum ranks.

Method	Unadjusted HR per pooled SD (95% CI), P	Adjusted HR per pooled SD (95% CI), P
DeSurv ($k = 3$)	1.42 (1.26–1.59), < 0.001	1.28 (1.11–1.47), < 0.001
Matched-budget unsupervised control ($k = 3$, $\alpha = 0$)	1.04 (0.94–1.16), 0.428	0.98 (0.88–1.09), 0.705
Rank-optimized unsupervised control ($k = 7$, $\alpha = 0$)	1.44 (1.30–1.59), < 0.001	1.40 (1.22–1.62), < 0.001

D1 retains prognostic information beyond established PDAC classifications

We first asked whether D1 simply recapitulated established PDAC subtype classifications. Across the pooled validation cohorts ($n = 570$, 388 events), binary PurIST and DeCAF classifications

together explained 34% of the within-dataset variance in the continuous D1 score. D1 nevertheless remained associated with survival after adjustment for both classifiers (HR per within-dataset SD 0.82, 95% CI 0.73–0.93, $P = 0.0014$). Adding D1 to a dataset-stratified Cox model with additive PurIST and DeCAF terms improved fit (partial likelihood-ratio $\chi^2 = 10.1$, 1 df, $P = 0.0015$). D1 therefore overlaps substantially with established tumor-intrinsic and stromal classifications while retaining prognostic information beyond those binary calls.

Because D1 spans tumor and stromal programs, we also asked whether it primarily reflected tumor purity. In the two datasets with purity estimates for all patients, D1 correlated only weakly with purity (Spearman $\rho = 0.06$ and -0.05), and adjustment for purity left the linear predictor’s survival association essentially unchanged (HR per SD: PACA-AU array, 1.44 versus 1.45; RNA-seq, 1.66 versus 1.65; Supplementary Table S7). Within the PACA-AU datasets, these findings argue against measured tumor purity as the main explanation for the observed associations, although they do not establish the cellular identity of D1’s restCAF-like component (Supplementary Information, Stromal resolution: which reference programs are recovered from bulk).

Method comparisons distinguish survival prediction from gene program recovery

We next asked whether comparable prognostic information could be recovered without survival-supervised factorization. At DeSurv’s rank, the matched-budget unsupervised control did not show a statistically significant survival association (Table 1).

The rank-optimized unsupervised control (the same pipeline at $\alpha = 0$ with rank selected by the same one-standard-error rule applied to its own Bayesian-optimization results, $k = 7$) carried comparable external prognostic information (Table 1; Supplementary Table S8). Reference-program coverage increased from 16 for DeSurv at $k = 3$ to 32 for the $k = 7$ control, while multiplicity remained low (1.00 versus 1.125; Supplementary Figs. S6 and S7), indicating broader reference-program coverage with little repeated mapping in the seven-factor control. Seven is where the one-standard-error rule landed for the unsupervised search, not a general minimum rank for comparable prognostic information. Survival supervision was not required to obtain a prognostic signal, but it altered which programs were identified and how that signal was decomposed.

We next asked whether D1 was specific to the DeSurv optimizer. We applied three other survival-supervised approaches to the training datasets (one supervised component each for supervised principal components and Cox partial least squares, one penalized fit for sparse Cox; Methods). All three produced patient scores strongly correlated with D1 (Pearson $|r| = 0.81$, 0.83 and 0.85 , respectively), whereas the leading unsupervised principal component was only weakly correlated with D1 ($|r| = 0.10$) and instead tracked the higher-variance D2/D3 structure (Supplementary Information, The D1 axis is not specific to DeSurv). This indicates a shared survival-associated direction in the patient scores, not that the three methods recover the same gene programs.

We then asked what DeSurv adds beyond recovery of this shared prognostic direction. Independently tuned supervised principal components and lasso-penalized Cox regression had pooled external C-indices of 0.60 and 0.60, versus 0.61 for DeSurv. For this comparison, DeSurv used per-program gene supports and unpenalized Cox coefficients re-estimated on the training scores, rather than the union-support predictor used for primary external validation. The selected genes of the supervised predictors, however, were enriched for basal-like tumor programs in every cross-validation seed and for none of the stromal, immune or exocrine programs examined, and their risk scores correlated only weakly with the separately resolved proCAF-like D2 program (lasso Cox, $|r| = 0.06$; supervised

PCA, $|r| = 0.14$; Fig. 3b). DeSurv places the shared prognostic direction within separately scored tumor and stromal programs and provides program-specific gene lists for biological investigation and future classifier development.

Survival supervision improves program recovery when prognostic signal contributes little expression variation

We used simulations with known latent structure to determine when survival supervision improves gene program recovery. Expression profiles were generated for $p = 3,000$ genes and $n = 200$ samples from $k = 3$ true programs, each with its own marker genes and all drawn from the same loading and activity distributions (Methods, Simulation studies). In the primary scenario, survival depended on the marker genes of one program only, and those genes contributed less transcriptomic variation than a shared background signal excluded from the survival-generating gene set. DeSurv and its unsupervised limit ($\alpha = 0$) were tuned by the same cross-validation procedure, giving a controlled comparison with and without survival supervision.

Under this setting, DeSurv more accurately recovered the genes defining the true prognostic program, whereas unsupervised NMF was more strongly influenced by the dominant background structure (Fig. 5a,b). Across 100 simulated datasets, median matched-factor precision, the proportion of the best-matching learned program’s selected genes that belonged to the true prognostic gene set, was 0.50 for DeSurv versus 0.07 for unsupervised NMF. Global marker recall, the proportion of the true survival-driving marker genes appearing in the union of all learned programs’ top-ranked gene sets, was 0.26 versus 0.15 (Supplementary Information). Median test-set C-index was 0.837 versus 0.724 ($\Delta = 0.113$, paired Wilcoxon $P < 0.001$), and DeSurv recovered the true rank ($k = 3$) more frequently (Fig. 5c).

Performance depended on how prognostic signal aligned with expression structure. In the mixed setting, the benefit of supervision differed between recovery of program-specific markers and recovery of shared background genes included in the survival-generating set (Supplementary Fig. S3). Under the null, test-set concordance was near chance for both methods. There, the selected supervision strength dispersed across its admissible range rather than concentrating at zero (median $\alpha = 0.59$, interquartile range 0.19–0.82 across 100 simulated datasets), consistent with weak discrimination among supervision strengths when concordance is near chance.

Discussion

DeSurv allows patient survival to shape gene-program discovery, yielding a tumor–stroma architecture that connects prognostic information with interpretable malignant and stromal programs. The practical value lies in what that architecture yields, program-specific gene lists that can be compared with established classifications and investigated biologically across cohorts. In PDAC, the architecture comprised a dominant survival-aligned program, D1, characterized by coordinated cross-patient variation in classical tumor and restCAF-like stromal features, alongside separately resolved proCAF-like stromal and basal-like tumor programs. D1 remained prognostic after adjustment for the established PurIST and DeCAF classifications, and was supported on its classical tumor side by GATA6 RNA in situ hybridization. The frozen programs then transferred without refitting across four independent external patient cohorts.

This difference in the recovered programs persisted when rank and tuning effort were matched. An unsupervised control at $\alpha = 0$, matched to DeSurv on rank, optimizer, tuning budget and consensus initialization and tuned on cross-validated concordance, did not recover D1. A rank-optimized unsupervised control recovered comparable prognostic information across a broader seven-factor representation, and conventional supervised predictors produced scores aligned with the dominant prognostic direction, correlating only weakly with the separately resolved proCAF-like program (Fig. 3b). These comparisons separate prognostic information from the representation in which that information is recovered.

This distinction also clarifies the role of the frozen gene-program basis. When programs are learned without outcome information, their prognostic relevance is assessed after discovery. In DeSurv, survival acts directly on the gene-program matrix W while the reconstruction objective requires the learned programs to continue representing the observed transcriptome. Once learned, the gene weights are held fixed, so external validation tests the same programs rather than a newly estimated representation. In CoxNMF and SurvNMF, the survival term depends on sample-side latent representations^{32,33}. In DeSurv, it depends directly on projections through the gene-program matrix, and new samples are scored using fixed gene weights without an additional sample-loading estimation step (Supplementary Table S1). Empirically, supervision reorganized rather than replaced the reconstruction-driven structure. D2 and D3 corresponded closely to the major stromal and tumor factors recovered by standard NMF, whereas D1 was poorly reconstructable from those matched-rank factors.

Program-level outputs are the practical product of this architecture, because the genes assigned to a program determine how it is interpreted and what is tested next. Previous PDAC studies illustrate how interpretable malignant and stromal programs can support biological characterization and the development of single-sample classifiers^{3-5,14}. Program-specific gene lists connect measured scores to biological annotation and provide candidates for subsequent classifier development. DeSurv extends this line of work by allowing patient outcomes to influence the gene weights themselves, and D2 remains available as separately measurable proCAF-like stromal context although the fitted risk score combines D1 and D3. Such outputs supply candidate signatures and hypotheses for biological investigation and biomarker development, without yet establishing clinical utility or a treatment-selection rule.

Rank choice and initialization can affect the recovered gene programs, and both were examined here. Standard NMF criteria gave conflicting rank choices in the PDAC training data, and increasing rank progressively broadened correspondence to established PDAC programs. The three-factor DeSurv solution and the seven-factor unsupervised control were selected by the same one-standard-error rule, and seven is the smallest rank within one standard error of the best configuration in the $\alpha = 0$ search, so the two differ in representation size rather than marking a minimum-rank requirement for either method. Survival supervision provides an outcome-informed criterion for shaping the representation, but it does not remove the non-uniqueness of matrix factorization, and individual DeSurv runs from different random starts recover appreciably different gene sets. Consensus fits built from disjoint halves of the same cached restart pool agreed closely with the reported basis (gene-loading correlation 0.97–0.99, top-270 Jaccard 0.73–0.88; Supplementary Information, Stability of the recovered gene programs). This supports stability to restart-pool composition within the training data, while leaving reproducibility under independent repetition, and the stability of the full consensus procedure under patient resampling, to be established.

The simulations identify conditions under which survival information improves program recovery. In the primary setting, the marker genes defining the prognostic program contributed less expression

variation than shared background genes excluded from the survival-generating gene set. DeSurv recovered those prognostic markers more accurately than its unsupervised limit. In the mixed setting, the benefit differed between program-specific markers and shared background genes, while test-set concordance was near chance for both methods under the null. These findings provide a controlled example of how survival supervision can prioritize prognostically relevant structure that contributes relatively little expression variation.

Three statistical qualifications bound the prognostic findings. DeSurv uses a proportional-hazards objective, while the external validation data showed attenuation of the prognostic association over follow-up (Results), so the reported hazard ratios are time-averaged summaries and extensions could accommodate non-proportional hazards or nonlinear survival relationships. The integrated Brier analysis recalibrated the score’s scalar slope and baseline hazard within each validation dataset, so it describes apparent prediction error rather than out-of-sample prediction error (Supplementary Information, Proper-scoring sensitivity of the external validation). The pooled training objective was fitted without a cohort term; within-cohort analyses argued against simple encoding of training-cohort membership, though cohort influence on the learned representation remains possible (Supplementary Information, Cohort handling in the training objective).

The question DeSurv addresses extends beyond PDAC. In any tumor type, the programs that dominate expression variation may differ from those most relevant to survival, so outcome information could help prioritize programs that reconstruction alone underweights. Testing generalizability will require multi-cancer evaluation of prognostic performance, program recovery and stability, with attention to cohort composition and event counts. Comparators should include unsupervised decompositions and strong penalized-regression baselines, which remain difficult to outperform consistently for out-of-sample prognosis³⁴. PDAC is the empirical demonstration presented here rather than evidence of pan-cancer validation.

In treated cohorts the frozen scores remained informative in exploratory analyses. In O’Kane/COMPASS, D1 remained prognostic in an analysis stratified by treatment arm, although arm-level sample sizes limited assessment of variation in the association across treatments, and the epithelial-enriched profiles support transport of its tumor-associated component in particular. The increase in the projected D2 score after treatment in the Linehan cohort indicates a shift toward proCAF-associated bulk expression, which may reflect altered cellular composition, transcriptional state or both, and does not establish a treatment-induced change in fibroblast state. Whether these programs predict differential treatment benefit therefore remains unresolved and will require adequately powered, prespecified treatment-by-program interaction analyses in studies designed to support treatment comparisons.

The tumor–stroma architecture described here is a bulk-tissue phenomenon. It reflects cross-patient covariation in tumor- and stroma-associated transcriptional programs rather than cellular co-expression or spatial adjacency within individual tumors. Within it, D1 is the dominant survival-aligned program rather than the architecture itself, and this organization is consistent with broader evidence that malignant lineage state and stromal composition vary together in PDAC^{35–37}. The restCAF-like and proCAF-like labels denote correspondence to reference signatures rather than direct resolution of single-cell CAF states. Adjustment for PurIST and DeCAF supports D1’s prognostic association beyond those classifiers, purity adjustment in the PACA-AU datasets leaves the risk score’s association essentially unchanged, and other supervised methods recover a related patient-score direction. GATA6 RNA in situ hybridization independently supports D1’s classical tumor component; these analyses do not establish the cellular identity of its stromal component. Single-cell or spatial profiling could clarify the cellular sources of these programs; linking those

sources to prognosis would additionally require outcome-annotated cohorts and scoring methods appropriate for the measurement platform.

In PDAC, incorporating survival during factorization recovered a three-program tumor–stroma architecture that transferred across independent cohorts while retaining biological components that could be interrogated separately. More generally, DeSurv provides a framework for asking not only whether transcriptomic data predict outcome, but which biological programs are recovered when outcome information helps shape their discovery.

Methods

Problem formulation and notation

Let $X \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p \times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k \times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{> 0}^n$ denote patient survival times, $\delta \in \{0, 1\}^n$ the event indicators ($\delta_i = 1$ for an observed event, $\delta_i = 0$ for censoring), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k \times n}$: for sample i with projected score $z_i = W^\top x_i$, the hazard is $h(t | x_i) = h_0(t) \exp(z_i^\top \beta)$, where h_0 is the baseline hazard. Defining factor scores by projection onto the learned gene programs ($Z = W^\top X$) rather than by the inferred loadings H lets the same gene weights score new cohorts without refitting, the property we exploit for external validation.

The DeSurv model

DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with survival outcomes. The method operates in a survival-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0, 1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W, H, \beta) = (1 - \alpha) \mathcal{L}_{\text{NMF}}(W, H) - \alpha \mathcal{L}_{\text{Cox}}(W, \beta), \quad (1)$$

where $\mathcal{L}_{\text{NMF}}(W, H)$ is the NMF reconstruction error (with an optional ℓ_2 penalty on H , set to zero in all analyses reported here) and $\mathcal{L}_{\text{Cox}}(W, \beta)$ is the elastic-net penalized partial log-likelihood. Both terms are normalized before weighting: the reconstruction error by $\|X\|_F^2$ and the partial log-likelihood by $n_{\text{event}}/2$. The W update then balances the magnitudes of the two gradients before α sets their relative contribution (Supplementary Information, Section 2). A key design choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\text{Cox}}$ is a direct function of W and β , and the survival gradient acts explicitly on the gene program matrix W . The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients^{12,38}. This construction makes clinical outcome part of program estimation rather than a post hoc filter applied to an unsupervised decomposition.

When $\alpha = 0$, DeSurv reduces to its unsupervised limit. The Cox coefficients β are then fit post hoc by penalized Coxnet on the projected scores $Z = W^\top X$ and are used only for scoring

and cross-validation; they do not affect the factorization. Note that the survival-tuned n_{top} still enters the consensus initialization of the final $\alpha = 0$ model (Supplementary Information, Bayesian Optimization). The Cox component also supports optional adjustment for sample-level covariates (for example, stage or grade); the analyses reported here use factor scores alone.

Optimization alternates updates for H , W , and β (Supplementary Information, Section 2), with backtracking used for proposed W updates. The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W 's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet-style cyclic coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ) . That coordinate descent operates on standardized scores and weights its quadratic approximation of the partial likelihood by $1/n_{\text{event}}$ rather than the $2/n_{\text{event}}$ of the objective, so the penalty it applies is effectively twice the nominal λ ; its proposal is accepted only if the objective does not increase, and β is refit on the converged W at the end. Convergence of the outer loop is assessed by the change in W and H between successive iterations, not by the change in the objective (Supplementary Information, Section 2). The β block is therefore an approximate rather than exact conditional minimizer; no reported λ or fitted model is affected, since the reported fits are those this procedure produces.

Because the implemented algorithm combines normalization, gradient balancing and a Coxnet approximation, we assessed optimization behavior empirically rather than relying on a formal convergence guarantee. Across independent random initializations of the reported PDAC model, all 100 runs terminated on the stopping criterion and the fitted objective decreased monotonically to numerically stable solutions (Supplementary Information, Empirical optimization stability). Because objective convergence does not imply that the same programs are recovered each time, we separately assessed program-level stability. Individual restarts are dispersed, whereas independently seeded applications of the reported consensus procedure reproduce the decomposition closely (Supplementary Information, Stability of the recovered gene programs). Closed-form update equations are given in Supplementary Information, Sections 1–2, and Bayesian-optimization control parameters in the Bayesian Optimization section.

Hyperparameter selection and cross-validation

Hyperparameters $(k, \alpha, \lambda, \xi, n_{\text{top}})$ were selected by maximizing the cross-validated Harrell C-index using Bayesian optimization (BO)^{26,27}. The final rank k was chosen by a one-standard-error rule: for each rank, the Bayesian-optimization configuration with the highest observed mean cross-validated C-index was retained, the threshold was set one fold-level standard error below the overall best configuration's mean, and the smallest rank whose retained configuration met the threshold was selected; the remaining hyperparameters were those of the overall best configuration (Supplementary Information, Bayesian Optimization). Because this rule combined $k = 3$ with the non-rank hyperparameters of the global optimum, a configuration not itself evaluated in the search, we examined a fixed $k = 3$ grid holding λ , ξ and n_{top} at those retained values as a sensitivity check: the nearest evaluated supervision level, $\alpha = 0.35$, gave a mean cross-validated C-index of 0.656 (SE 0.010), above the one-standard-error threshold of 0.644, and applying the same one-standard-error rule to that grid also selects $k = 3$ (Supplementary Information, Bayesian Optimization and Cross-validated C-index across factorization rank). All model selection was performed entirely within the training data using cross-validation; no validation cohort data were used during any stage of tuning, and out-of-sample generalization was assessed only on the external validation cohorts. Each fold was trained using multiple random initializations, and fold-level performance was defined as the

average C-index across initializations; the consensus initialization described next was used only for the final model and is not reproduced within each fold.

The consensus initialization aggregates multiple DeSurv runs into a gene-gene co-occurrence matrix and constructs an initialization W_0 from the resulting clusters (Supplementary Information, Consensus initialization for the final model).

Before projection, the gene support was restricted to the union of each program’s BO-selected top genes, ranked by factor specificity (Supplementary Information, Section 8), giving the truncated matrix $\widetilde{W}$ that was held fixed for all validation cohorts.

Two scoring conventions follow from this support and are used for different quantities. Factor-level scores retain all k columns, $Z = \widetilde{W}^\top X_{\text{new}}$, and are used for the per-factor and pooled survival analyses. Where a single risk score is required, as in cross-validated tuning, the columns are first collapsed to $\theta = \widetilde{W}\beta$ and θ is scaled to unit ℓ_2 norm before projection; this normalization changes the resulting scalar risk score only by a positive multiplicative constant and therefore leaves concordance unchanged. The normalization applies to the collapsed score rather than to individual columns of W . Here, projection denotes multiplication by the fixed truncated gene-weight matrix, $\widetilde{W}^\top X_{\text{new}}$; because truncation does not renormalize the columns, these scores are not orthogonal geometric projections. During model fitting, β was estimated from the full-support scores $W^\top X$ over all analyzed genes. For external scoring, the fitted β was carried forward unchanged and applied to scores computed on the union support $\widetilde{W}$ defined above; β was not refit after truncation. Cross-validation evaluated this same truncation-and-scoring construction when selecting n_{top} .

In PDAC, BO selected $n_{\text{top}} = 270$ genes per factor. The ridge penalty on the loadings (H) was held fixed at 0 in all analyses and was not tuned; factor scaling was instead controlled directly by column-normalization of W during optimization.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance (cross-validated C-index) rather than the unsupervised diagnostics commonly used for NMF rank selection^{13,28}. In the PDAC training data, standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks, a known limitation of unsupervised rank-selection criteria that can disagree on the same dataset (Supplementary Fig. S5). Cross-validated concordance varied little across candidate ranks, with no clear optimum anywhere between $k = 2$ and $k = 12$ and a total spread across that range of about two standard errors (Supplementary Fig. S4). The one-standard-error rule applied to the rank-wise BO results accordingly selected the parsimonious $k = 3$; the final model retained $\alpha = 0.334$ and the other non-rank hyperparameters from the overall best configuration. Biological support for the three-program solution comes from reference-program correspondence and the GATA6 comparison rather than from a sharp statistical optimum.

External validation cohorts were scored by projecting new datasets onto the learned programs via $Z = \widetilde{W}^\top X_{\text{new}}$, and survival associations were evaluated using Harrell’s C-index and hazard ratios. Harrell’s C-index is rank-based and assesses discrimination only, not calibration or time-dependent prediction error, so we additionally report a proper score as a sensitivity check. This is the integrated Brier score and its index of prediction accuracy relative to a Kaplan–Meier reference (Supplementary Information, Proper-scoring sensitivity of the external validation).

Because the survival-supervised information lives in W rather than H , scoring is direct multiplication by the fixed gene-weight matrix, $\widetilde{W}^\top X_{\text{new}}$, requiring no retraining or validation survival information. Related supervised-NMF methods instead place the survival term on the sample side: CoxNMF maximizes a Cox partial likelihood whose covariates are the columns of H ³², and SurvNMF jointly

learns the factorization and a Cox model on the sample-side latent matrix, which its own transposed notation writes as W^{33} . Because the risk score under either formulation is a function of the sample loadings rather than of the gene-program basis, scoring an external cohort requires re-estimating those loadings; SurvNMF obtains them for held-out samples by a constrained least-squares fit (Supplementary Table S1). Validation survival outcomes were therefore used only for downstream performance evaluation, so reported external hazard ratios reflect purely out-of-sample generalization. Further training and validation details appear in Supplementary Information, Part I.

To characterize the biological identity of the learned factors, we quantified how strongly each factor’s gene loadings aligned with independently defined PDAC tumor and microenvironmental gene programs. For each factor, we correlated its gene loadings with binary membership indicators for each reference program (Spearman correlation computed over the union of the top 50 genes of all factors of that method, intersected with the reference-program gene universe; this differs from the factor-to-factor loading correlations, which use all analyzed genes), annotating Benjamini–Hochberg-adjusted P values within each factor as descriptive marks, because the gene universe is itself selected by loading. The same analysis was applied to DeSurv and to standard NMF fit at the same rank. For display, reference programs whose Spearman correlation exceeded 0.2 with at least one factor of either method are shown, as the same row set in both panels of the main figure; the same threshold defines correspondence in the coverage analysis (Supplementary Information, Reference-program coverage, edges and multiplicity).

The matched-rank comparator shown in the figure is Lee multiplicative NMF (`NMF::nmf` with `nrun = 30`, which returns the single run of the thirty with the lowest reconstruction residual, followed by a post hoc Cox fit on its factor scores), which does not receive DeSurv’s consensus initialization or hyperparameter tuning. We therefore also fit a matched-budget unsupervised control, an unsupervised factorization control with outcome-based hyperparameter tuning: at $\alpha = 0$ survival does not enter the factorization objective, but cross-validated survival performance still tunes the remaining hyperparameters and n_{top} , and every other component of the DeSurv procedure is held fixed. Bayesian optimization was run with α pinned to 0 and the rank pinned to DeSurv’s selected k , using the same optimizer, search budget, gene filter and cross-validation folds. The selected model was then fit by the same 100-restart consensus procedure. This control isolates removal of survival supervision at the pipeline level, with the remaining hyperparameters re-optimized rather than held numerically fixed. A further sensitivity analysis instead held DeSurv’s own λ , ξ and n_{top} fixed and varied only α . Both controls are reported in Supplementary Information, Matched-budget unsupervised control at the same rank.

To corroborate the classical component of D1 with an independent assay, we evaluated the association between projected D1 scores and GATA6 RNA in situ hybridization measurements in the COMPASS cohort among the 33 tumors with both measurements, using Spearman correlation. Because *GATA6* is itself in the 810-gene projection support (805 of these genes were measured in the COMPASS expression data and contribute to the projected scores), part of this correlation could reflect the score being built from the assayed gene. We therefore repeated the comparison with *GATA6* removed from the projection, and the correlation was essentially unchanged (Spearman $\rho = 0.67$ versus 0.68), indicating the association is not an artifact of self-correlation.

Supervised comparator analyses

To distinguish prognostic discrimination from biological representation, we independently fit supervised principal components (`superpc`; cross-validated screening threshold and component number)

and lasso-penalized Cox regression (`glmnet`; $\lambda_{\min}$) using the same 1,970-gene training universe as DeSurv. Each method performed feature selection and tuning using the training data only, after which the fitted model was frozen and applied to rank-transformed validation cohorts without refitting.

The two comparators select features by different mechanisms, and neither matches DeSurv’s. The penalized Cox comparator uses a pure lasso penalty (`glmnet` with mixing parameter 1, distinct from DeSurv’s supervision parameter α), whereas supervised principal components applies no penalty at all, instead screening genes by a cross-validated feature-score threshold and extracting a cross-validated number of components. DeSurv’s internal Cox step uses an elastic net whose Bayesian-optimization-selected mixing parameter ($\xi = 0.056$) sits close to ridge. Both comparators retain far fewer genes than DeSurv’s 810-gene support (across resampling seeds, sparse Cox selected a median of 14 genes, range 9–33, and supervised principal components 39, range 5–222), and both were tuned by their standard method-specific cross-validation rather than a budget matched to DeSurv; the comparison therefore characterizes the representations each approach recovers rather than benchmarking tuned predictive performance.

No matched-budget control was run for these supervised comparators, as was done for the unsupervised arm. Correspondence between each supervised risk score and the DeSurv programs D1–D3 was quantified in the training cohort using absolute Pearson correlation. Additional analyses of selected-gene enrichment and compartment attribution are described in the Supplementary Information.

Simulation studies

Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions in three scenarios: a prognostic scenario in which the factor-specific survival signal contributes little transcriptomic variation relative to the shared background that is excluded from the survival-generating gene set, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant signals partially overlap. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\text{Gamma}(\text{shape} = 3, \text{rate} = 0.8)$ and cross-loadings onto other factors from $\text{Gamma}(1, 20)$; background gene loadings from $\text{Gamma}(2, 1)$ across all factors; and noise gene loadings from $\text{Gamma}(1, 20)$. Sample-level factor activities H were drawn from $\text{Gamma}(2, 2)$. Each dataset comprised $p = 3,000$ genes, $n = 200$ samples, and $k = 3$ factors, with 150 marker genes per factor and 500 shared background genes. The three factors are exchangeable in this design: the prognostic factor is not generated with lower variance than the other two. The variance-dominant signal excluded from the survival-generating gene set is the shared background block, which loads on every factor, so the factor-specific marker genes that carry survival contribute less of the total expression variation than that block. Because the background genes load on the same latent activities as the markers, their expression remains correlated with the linear predictor; “background” here denotes exclusion from the survival-generating gene set, not statistical independence of outcome. The expression matrix actually analyzed was not the noise-free product WH : a per-gene baseline drawn as $\mathcal{N}(3, 1)$ was added to every row, independent Gaussian noise with standard deviation 0.01 was added to every entry, and negative values were clipped to zero. Cox regression coefficients

were $\beta = (2, 0, 0)^\top$ in the prognostic and mixed scenarios and $\beta = 0$ in the null scenario; the mixed scenario differs from the prognostic scenario only in its survival gene set, 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, which creates partial overlap between the prognostic signal and the variance-dominant background.

Survival times were generated from an exponential distribution, from the noise-free expression only. Let $X_0 = WH$ denote that noise-free matrix and $\widetilde{W}$ the survival-generating weight matrix, a masked copy of the column-normalized W that retains, in each column, only the loadings of that factor’s survival-generating genes and is zero elsewhere: the 150 markers of that factor in the prognostic and null scenarios, and 150 markers plus 150 background genes in the mixed scenario. True factor scores were $Z_{\text{true}} = X_0^\top \widetilde{W} \in \mathbb{R}^{n \times k}$, standardized per factor across all $n = 200$ samples before the 70/30 train/test split; the linear predictor was $\eta = Z_{\text{true}}\beta$, so with $\beta = (2, 0, 0)^\top$ only factor 1 contributes; event times were drawn as $T \sim \text{Exponential}(\lambda_0 e^\eta)$ with $\lambda_0 = 0.05$, and censoring times independently as $\text{Exponential}(0.02)$. Each scenario comprised 100 replicates, each with a held-out test split. Each dataset was analyzed using DeSurv and unsupervised NMF ($\alpha = 0$, the unsupervised limit of the same pipeline) followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across replicates using test-set concordance and matched-factor precision of prognostic-gene recovery.

Matched-factor precision was quantified separately for each survival-driving factor. We first matched each true program to the learned factor whose top-ranked genes had the highest precision against it. Precision was then the proportion of that learned factor’s top- n_{top} genes belonging to the ground-truth prognostic gene set, with genes ranked by factor-specificity score and n_{top} set to the Bayesian-optimization-selected signature size. Because the matching selects the factor by the precision that is then reported, matched-factor precision is optimistic by construction. We also report global marker recall, the fraction of the true survival-gene set recovered anywhere in the union of all learned factors’ top- n_{top} gene sets; it is insensitive to factor matching, although, like precision, it depends on each method’s selected n_{top} (Supplementary Information). In the primary scenario the target comprised the 150 factor-specific marker genes driving survival ($\beta = (2, 0, 0)$); in the mixed scenario, marker-only and full survival-gene definitions of the target were evaluated separately.

Real-world datasets

We analyzed publicly available and controlled-access RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Eligible samples were primary PDAC tumors with a measured bulk expression profile and valid overall-survival annotation, subject to cohort-specific quality-control and exclusion criteria detailed in Supplementary Information, Section 6; accrual and follow-up periods are as defined in each cohort’s original publication. Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. Within each training cohort, genes were ranked separately in descending order of mean expression and of variance, and the 3,000 genes with the smallest sum of the two ranks were retained, selecting genes that are both highly expressed and highly variable. The intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. No missing-data imputation was performed: all analyses used complete cases, and analyses adjusted for the PurIST and DeCAF classifiers were restricted to samples with available classifier calls. Of the seven PDAC expression datasets we

considered, two were used for training (TCGA and CPTAC) and five for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo); because the two PACA-AU platforms profile one patient cohort, these five validation datasets span four independent patient cohorts. To harmonize differences in scale across cohorts, filtered gene expression data were within-sample rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility). Within-sample ranking reduces platform- and library-size differences in expression scale that would otherwise dominate cross-cohort projection. The training Cox objective was fit to the pooled TCGA and CPTAC samples without a cohort term; cross-validation folds were constructed within cohort with stratification by event status. Both training cohorts are RNA-seq, were rank transformed within sample, and had similar event rates (52% in TCGA-PAAD and 50% in CPTAC; Supplementary Table S2); all validation analyses stratify by dataset, and a descriptive check against cohort-driven association is given in the Supplementary Information (Cohort handling in the training objective). More details about the datasets can be found in Supplementary Information, Section 6.

Per-cohort participant flow is summarized in Supplementary Table S2. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, missing tumor grade (an inclusion criterion of the curated training set, not a model input), or quality-control failure, yielding 273 analytic training samples (TCGA-PAAD 144, CPTAC-3 129; 139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation profiles across five platform datasets: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181).

PACA-AU included partially overlapping array and RNA-seq datasets (46 shared patients). Platform-specific analyses retained each dataset separately; for analyses combining validation datasets, duplicate patient accessions were represented once, preferentially retaining the RNA-seq profile and excluding the corresponding array profile, consistent with the preprocessing rule used in the previously published DeCAF analysis⁵. This yielded a combined validation population of 570 unique patients (388 deaths) spanning four independent patient cohorts.

Ethics, sex and gender, and ancestry

Each dataset was collected under the ethical approvals and consent procedures of the original study. This secondary analysis used only de-identified, previously collected data and was not submitted for separate Institutional Review Board (IRB) review at the University of North Carolina at Chapel Hill. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies) and our access was limited to the de-identified expression and survival data they provide. Sex was recorded in every cohort and was included as a covariate in the clinical-adjustment analyses (Supplementary Table S7), where the prognostic association of the DeSurv linear predictor was unchanged. We did not conduct sex-stratified analyses, because the study was not powered for subgroup comparisons within individual cohorts, and we did not stratify by race or ethnicity because those annotations were not consistently available across cohorts. Whether the prognostic relevance of these gene programs differs by sex, race, or ethnicity remains an important question for adequately powered, more comprehensively annotated cohorts.

Treated-cohort analyses

Two independently treated PDAC cohorts were used only for projection of the frozen programs (Supplementary Information, Translational transportability). Projection used the frozen union support without any retraining: expression was rank transformed within sample over the genes each cohort shares with the trained gene-program namespace, support genes absent from a cohort were omitted from that cohort’s projection rather than imputed (805 of the support genes were measured in the COMPASS data), and the resulting program scores were standardized within each treated cohort before analysis. In the O’Kane/COMPASS cohort of treated metastatic PDAC ($n = 173$ patients, 123 deaths; laser-capture microdissected, epithelial-enriched expression), the association of the projected D1 score with survival was reported both as arm-specific estimates and as a common slope from a Cox model stratified by treatment arm, which allows arm-specific baseline hazards, with a treatment-by-D1 interaction test reported descriptively; no estimate is pooled across cohorts. In the Linehan cohort of borderline-resectable or locally advanced disease, 29 patients with paired pre- and post-treatment biopsies were compared on the projected D2 score by a paired Wilcoxon signed-rank test; this analysis concerns program level, not survival. Both analyses were exploratory and not preregistered.

Use of large language models

During preparation of this manuscript, the authors used large language models (Claude, Anthropic; ChatGPT, OpenAI) for language editing, for assistance with code used to generate and format the display items, for verification of the written algorithm description against the released code, and for critical review of the manuscript’s presentation and evidentiary framing. All model output was directed, reviewed, and verified by the authors, who take full responsibility for the work. The models were not used to design the study or to develop the DeSurv methodology, and they do not meet the criteria for authorship.

Data Availability

The training and validation datasets used in this study are publicly available or accessible through controlled-access repositories. Training data were obtained from the Genomic Data Commons: TCGA-PAAD³⁹ and CPTAC-3⁴⁰. Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830⁴¹; Puleo, E-MTAB-6134⁴²), the Gene Expression Omnibus (GEO; Moffitt, GSE71729⁴³), and the International Cancer Genome Consortium (ICGC): the PACA-AU array cohort is publicly available from the Gene Expression Omnibus (GSE36924), and the PACA-AU RNA-seq cohort from the European Genome-phenome Archive (EGA) study EGAS00001000154 under controlled access⁴⁴. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study⁵; cohort details and sample sizes after quality control are summarized in Supplementary Information, Section 6. The Linehan FOLFIRINOX $\pm$ CCR2-inhibitor series, used for the paired pre- and post-treatment comparison (Fig. 4b), and the O’Kane/COMPASS treated metastatic PDAC cohort used both for the D1 prognostic-transportability survival analysis (Fig. 4a) and, in the subset of 33 tumors with matched staining, for the GATA6 RNA in situ hybridization comparison (main-text Fig. 2e), are restricted-access clinical-trial datasets that are not publicly deposited; raw data are available from the respective study investigators under a data-use agreement, subject to the governing trial consents and institutional approvals. To preserve

reproducibility without redistributing restricted data, the derived per-cohort summary statistics reported in the main text and Supplementary Information are provided in the reproducibility repository (file `results/treated_cohort_stats.rds`), so that every treated-cohort value can be inspected and verified. Code used to generate the COMPASS D1–GATA6 comparison (main-text Fig. 2e) is also provided in the reproducibility repository; reproduction of the patient-level analysis requires authorized access to the COMPASS data.

Code Availability

All analyses were performed in R (version 4.6.0; R Core Team) using DeSurv (v1.0.1; this work), NMF (v0.28)³⁸, glmnet (v5.0)²³, survival (v3.8.6), ggplot2 (v4.0.3) and cowplot (v1.2.0); Bayesian optimization for hyperparameter selection was implemented within DeSurv. The DeSurv R package is available at <https://github.com/rashidlab/DeSurv> (archived at Zenodo, DOI 10.5281/zenodo.20150929). Reproducibility code, processed data, and the exact session-info file are available at <https://github.com/rashidlab/DeSurv-paper> (archived at Zenodo, DOI 10.5281/zenodo.20150946); this includes `code/11_treated_cohort_analysis.R`, which generates the treated-cohort summary statistics above.

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Author Contributions

Conceptualization: A.M.Y., N.U.R. Methodology: A.M.Y., A.Y., D.L., N.U.R. Software: A.M.Y. Formal analysis: A.M.Y. Investigation: A.M.Y. Data curation: A.M.Y., X.L.P. Visualization: A.M.Y. Writing – original draft: A.M.Y., N.U.R. Writing – review and editing: A.M.Y., A.Y., X.L.P., J.J.Y., D.L., N.U.R. Resources (PDAC domain expertise and biological interpretation): X.L.P., J.J.Y. Supervision: N.U.R. Funding acquisition: A.M.Y., N.U.R. All authors reviewed and approved the final manuscript.

Competing Interests

N.U.R. and J.J.Y. are named as inventors on US patents US20220127676A1 and US20250066856A1, which relate to pancreatic cancer subtype classification. The PurIST and DeCAF classifiers were used only as published, independently derived comparators: their subtype calls were obtained with the released classifiers and enter only as fixed covariates in the adjustment analyses, not as components of DeSurv. The other authors declare no competing interests.

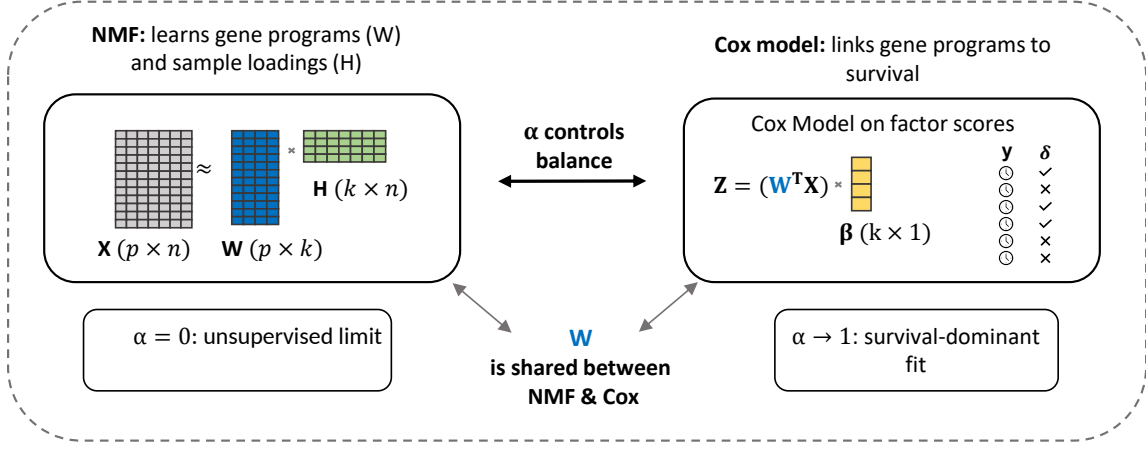
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a The DeSurv model



b Data-driven model selection via Bayesian optimization

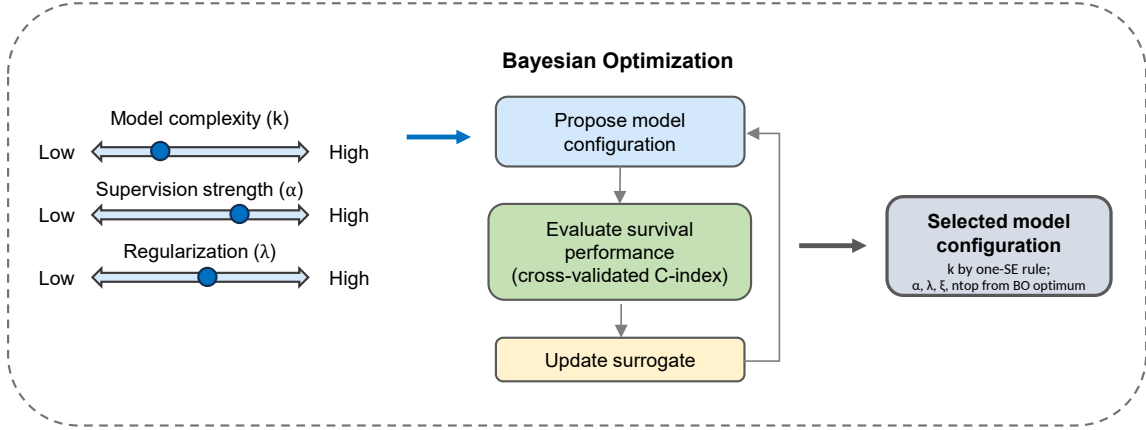


Figure 1: Overview of the DeSurv framework. (a) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^T X$ serve as covariates in the Cox model with coefficients β . A supervision parameter $\alpha \in [0, 1]$ controls the balance between reconstruction fidelity ($\alpha = 0$, the unsupervised limit) and survival prediction (α close to 1). (b) Hyperparameters (rank k , supervision strength α , regularization λ , ξ and n_{top}) are searched by Bayesian optimization over cross-validated concordance; the rank is then chosen by a one-standard-error rule and the remaining hyperparameters are retained from the best configuration (Methods). Survival supervision acts on the gene-program matrix W during discovery, rather than being applied only after an unsupervised factorization has been completed; following training-only model selection, the resulting gene weights are fixed for direct evaluation in external cohorts.

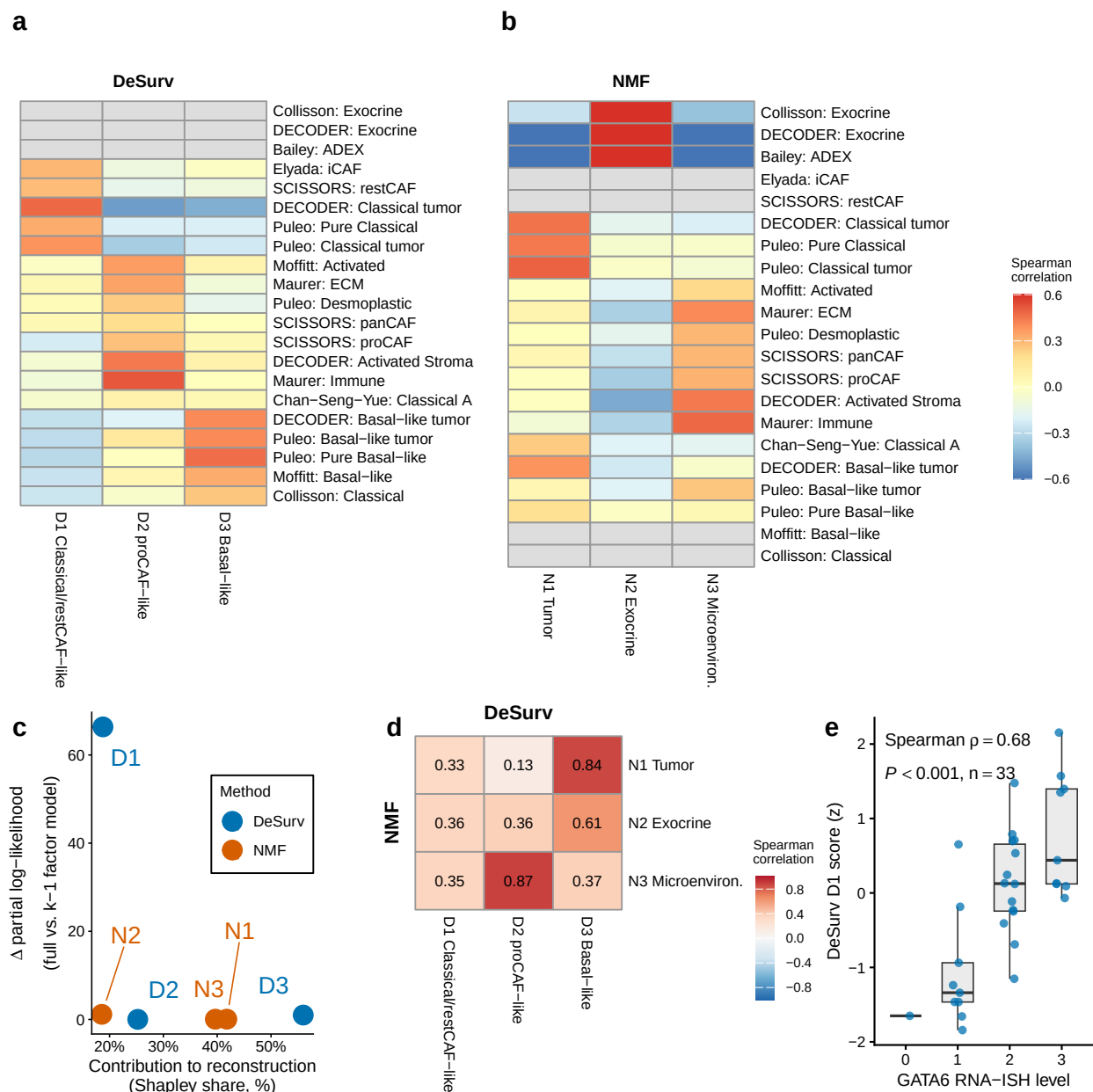


Figure 2: Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (a,b) Spearman correlations between factor gene rankings (top 50 genes per factor) and established PDAC gene programs for DeSurv (a) and standard NMF (b) at $k = 3$. Both panels display the same rows, the union of reference programs correlating at Spearman $\rho > 0.2$ with at least one factor of either method; grey cells denote programs with none of their genes in that method's displayed gene space, where the correlation is not computable. The rows labeled SCISSORS: restCAF and SCISSORS: proCAF display the SCISSORS iCAF and myCAF gene sets under the naming used by the DeCAF classification (Supplementary Information). These rows are correspondences between bulk gene programs and reference signatures, not direct identification of CAF cell states. (c) Per-factor reconstruction contribution (normalized Shapley share, summing to 100%) versus survival contribution, the decrease in the maximized unpenalized Cox partial log-likelihood when one factor is dropped from the three training-cohort scores projected over all analyzed genes and the Cox model is refitted using the remaining scores. These increments are conditional and nonadditive; D1 was estimated using these survival outcomes, so its increment locates the survival signal in the fitted model rather than testing whether that signal is real. (d) Pairwise correspondence between NMF and DeSurv factors (Spearman correlation of W-matrix gene loadings over all genes). (e) DeSurv D1 score versus GATA6 RNA in situ hybridization level in the COMPASS cohort (Spearman correlation: $n = 33$).

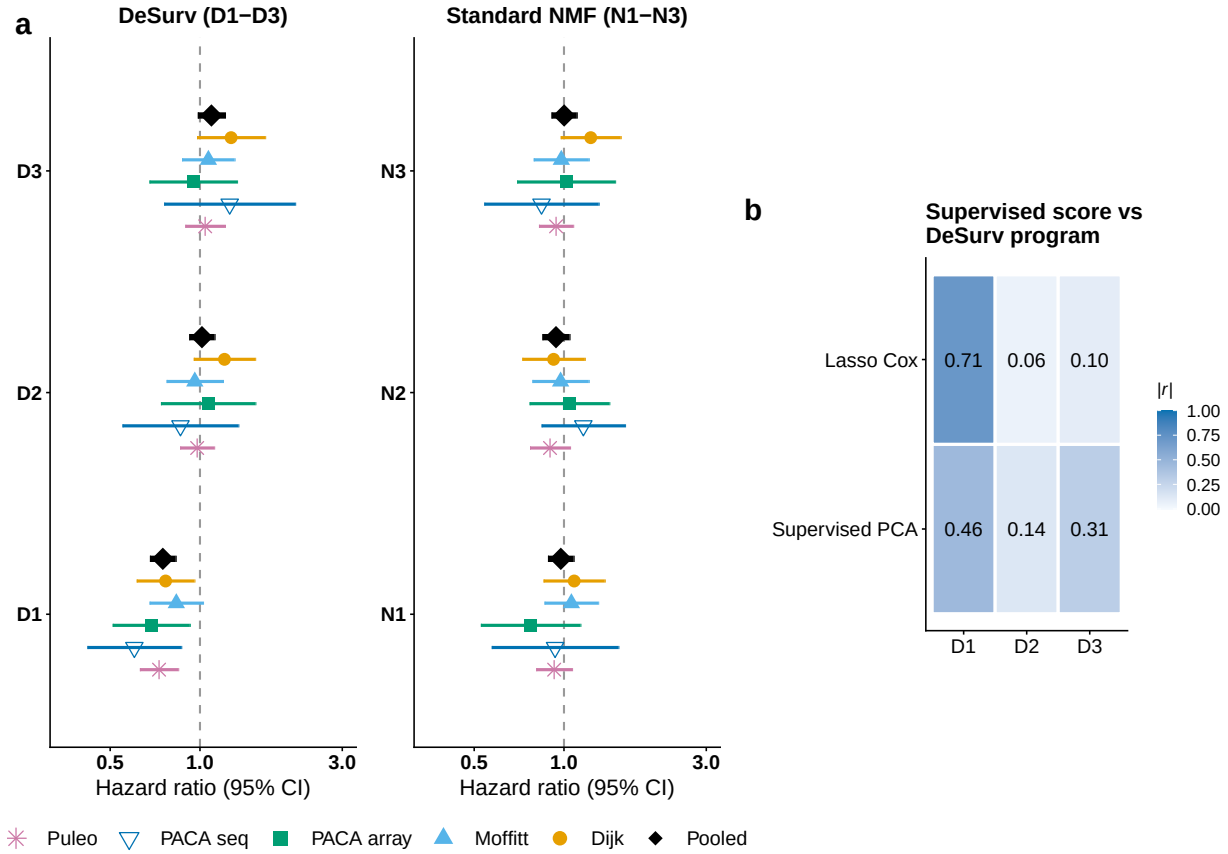


Figure 3: Survival-aligned programs generalize across independent PDAC patient cohorts (five external validation datasets: Dijk, Moffitt GEO array, PACA-AU array, PACA-AU RNA-seq and Puleo; combined $n = 570$ unique patients, 388 deaths, with PACA-AU patients profiled on both platforms represented once via RNA-seq). (a) Forest plot of per-dataset univariate hazard ratios (95% CIs) for each DeSurv program (left) and standard NMF factor (right); the platform-specific PACA-AU array and RNA-seq estimates are shown separately, and the black diamonds denote the combined estimate over unique patients (per-factor dataset-stratified Cox on the de-duplicated patient-level data, with each program standardized within dataset and scored on the union top-gene support; PACA-AU array duplicates excluded). Puleo is the largest contributor, supplying 288 of the 570 unique patients (51%), which is why the combined estimate is also reported with each dataset omitted in turn; those leave-one-dataset-out estimates range from 1.34 to 1.45. (b) Correspondence ($|\text{Pearson } r|$) of tuned supervised risk scores (lasso Cox and supervised PCA) with the three DeSurv programs (D1–D3) in the training data; supervised scores concentrate on D1 and are only weakly correlated with the D2 stromal program. These score correspondences do not establish recovery of identical gene programs. No validation-cohort survival outcomes were used in training, scoring or hyperparameter selection.

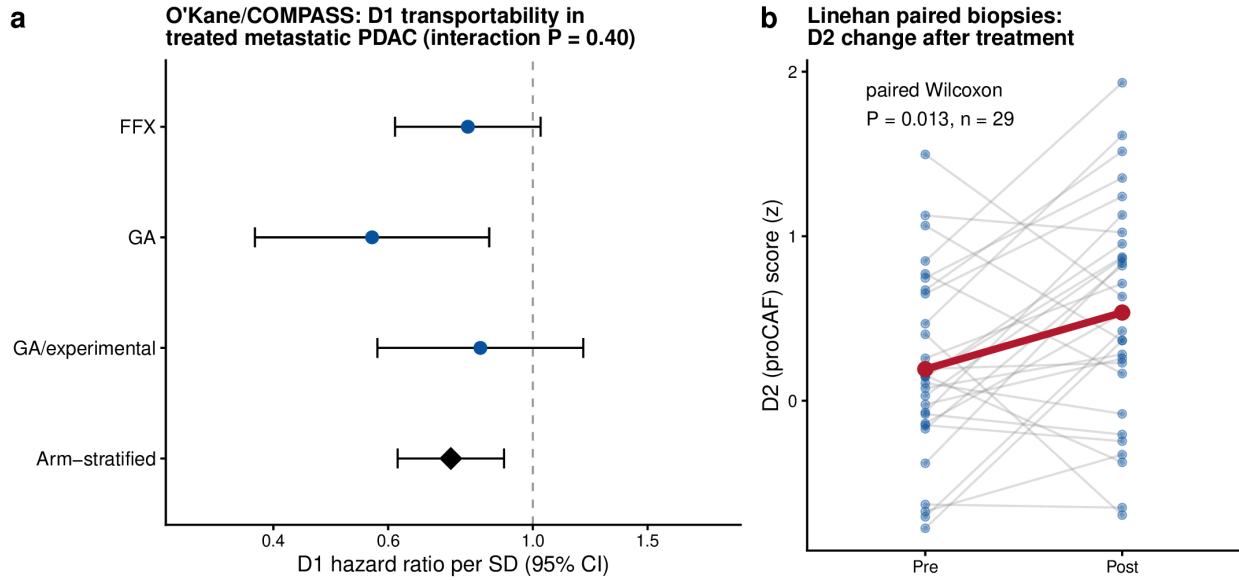


Figure 4: Transportability of the frozen DeSurv programs into independent treated PDAC cohorts. (a) Prognostic transportability of D1 in treated metastatic PDAC. Forest plot of hazard ratios per SD of the frozen D1 score within each O'Kane/COMPASS treatment arm (FOLFIRINOX, gemcitabine/nab-paclitaxel [GA], and GA/experimental), scored from laser-capture microdissected, epithelial-enriched profiles; the diamond is the common association estimated across arms from a stratified Cox model with treatment-arm-specific baseline hazards, HR 0.75 (95% CI 0.62–0.90, $P = 0.003$). D1 was protective in all three arms; the arm-by-score interaction was not significant (interaction $P = 0.40$), though the individual arms are underpowered for detecting heterogeneity. Error bars denote 95% confidence intervals. (b) The proCAF program D2 before and after treatment in paired Linehan biopsies ($n = 29$, paired Wilcoxon $P = 0.013$); grey lines are individual patients, the red line the group mean (+0.34 SD). Because treatment, stage and cohort were not independently varied, these analyses support prognostic and biological transportability rather than treatment-predictive effects.

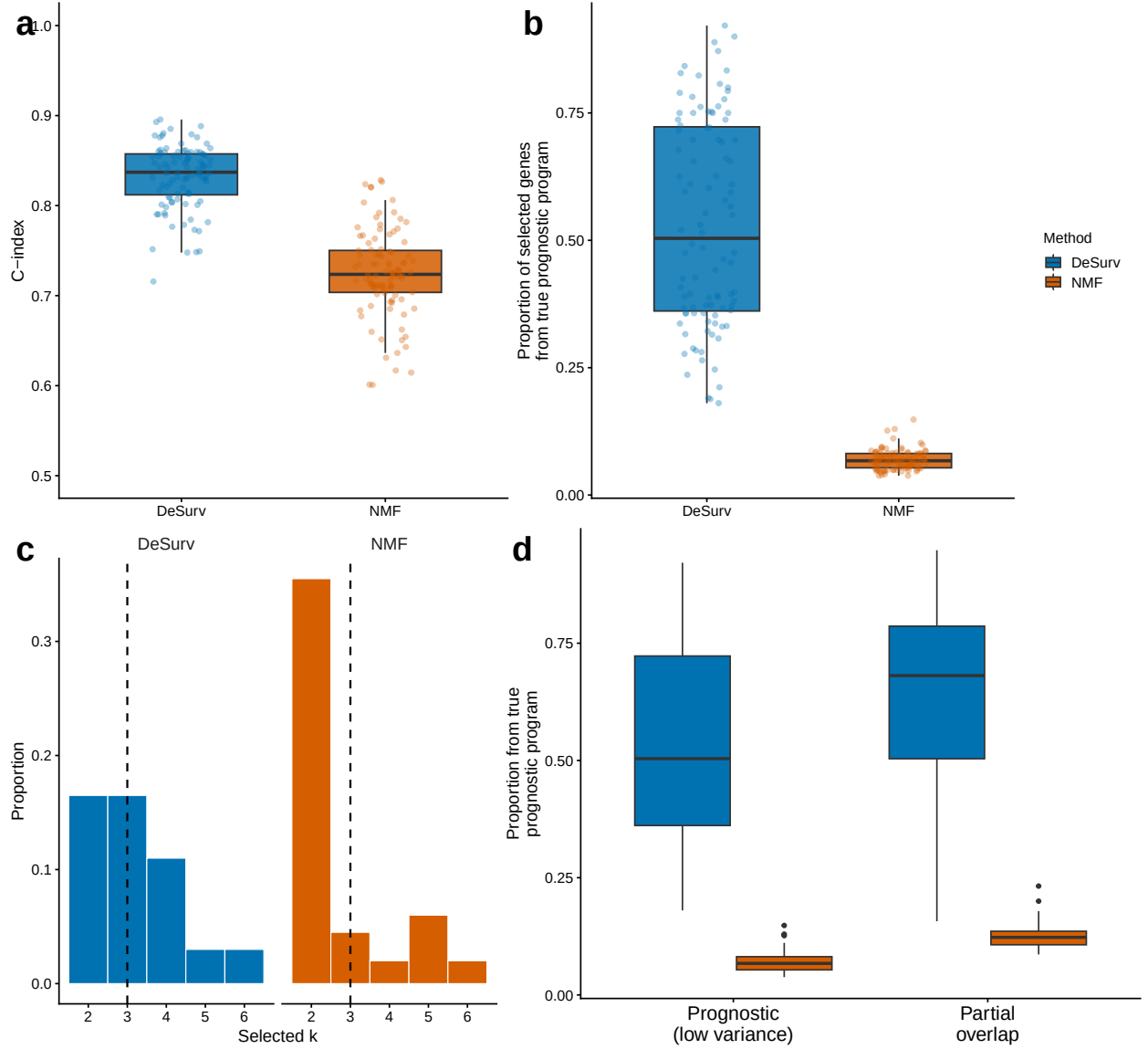


Figure 5: Survival supervision improves recovery of prognostic gene programs when survival signal is not variance-dominant. (a) Test-set concordance across 100 simulated datasets comparing DeSurv with unsupervised NMF ($\alpha = 0$; $p = 3,000$ genes, $n = 200$ samples, true $k = 3$); medians 0.837 versus 0.724 (paired Wilcoxon $P < 0.001$). (b) Recovery of genes defining the true prognostic program, quantified as the proportion of selected genes in the best-matching learned program that belonged to the ground-truth prognostic gene set; medians 0.50 versus 0.07. The best-matching program is selected using the known simulated gene set; this evaluates recovery with access to ground truth. Global marker recall, which uses no matching, is reported in the Supplementary Information. In (a) and (b), boxes show median and interquartile range; whiskers extend to $1.5 \times \text{IQR}$; points are individual simulated datasets. (c) Distribution of selected ranks across simulated datasets for DeSurv versus unsupervised NMF. (d) The same matched-factor precision metric as in (b), computed for two signal regimes: a prognostic program whose genes carry low transcriptomic variance relative to the shared background excluded from the survival-generating gene set (separated from the dominant variance) versus one that partially overlaps the dominant variance. DeSurv recovers the prognostic program more accurately in both regimes, whereas the unsupervised solution is more strongly influenced by the dominant expression structure; the null scenario (no prognostic program) is shown in the Supplementary Information. The concordance gap in (a) is not an expected discrimination gain in real data, where the two approaches perform comparably (Table 1).